Vaidik Shah

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Education

California State University, Long Beach **May 2027**
Master of Science in Computer Science *Long Beach, CA, USA*
Key Courses: Artificial Intelligence, Machine Learning, Distributed Computing, Natural Language Processing

Gujarat Technological University **June 2024**
Bachelor of Science in Computer Science *Ahmedabad, India*
Key Courses: Object Oriented Programming, Data Structures, Relational Database, Computer Networks, System Design

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, C++, C#, Golang, Rust
Backend & Databases: Node.js, Express.js, SQL (MySQL, PostgreSQL), NoSQL (MongoDB, DynamoDB)
AI/ML: PyTorch, TensorFlow, Scikit-learn, XGBoost, R, LangChain, NLP, Computer vision, NumPy, Pandas, Tableau, SAS
Cloud/DevOps & Web: AWS, GCP, Azure, CI/CD, Git, Linux/Unix, React, Angular, Vue, Bootstrap, Web3, HTML5/CSS
Software Engineering Fundamentals: Algorithms, Debugging, Scalable Systems, SDLC, Multi-threaded Programming, Agile

Experience

Technolee **July 2024 - July 2025**
Software Engineer (Data & ML) *Ahmedabad, India*

- Engineered and deployed an **LSTM-based RUL prediction model** on 100K+ run-to-failure vibration samples, **reducing prediction error by 22%** and **cutting projected unplanned downtime by 15% (12 hrs/month)**.
- Built and deployed a **real-time gaze estimation pipeline** on **NVIDIA Jetson Orin Nano** at **22 FPS**, integrating face detection and 3D gaze estimation models trained on 1.4M+ images via a REST API, **automating audience analytics for 10+ billboard locations.**
- Designed **custom signal preprocessing and feature extraction pipelines** for high-frequency vibration sensor data, **reducing preprocessing time by 68%**, enabling faster model iteration and scalable deployment of the RUL prediction system.

Indian Space Research Organization (ISRO) **January 2024 - July 2024**
AI/ML Engineer Intern *Ahmedabad, India*

- **Built enterprise satellite data pipeline** with automated validation protocols, **achieving 99.9% data integrity,** while processing 1+GB of datasets daily for solar insolation forecasting in renewable energy applications.
- Optimized **LSTM-based forecasting models using PyTorch/CUDA and MATLAB**, leveraging **advanced statistical techniques** to **reduce prediction error by 25%** and **improve model efficiency by 20%.**
- Developed **data visualization dashboards** using **Matplotlib and Seaborn**, transforming complex satellite data into actionable insights and **reducing reporting cycles by 40%.**

Cre-Art Solutions **June 2023 - August 2023**
Software Engineer Intern *Ahmedabad, India*

- **Redesigned Django RESTAPIs** with advanced OOP patterns, achieving a **20% performance boost** while processing **50K+ daily financial transactions** for a trading platform.
- Built an **automated validation framework** with **real-time error detection**, reducing **accounting discrepancies by 18%** and ensuring financial compliance standards across high-volume transaction processing.
- Built interactive **data visualization dashboards in Power BI with secure SQL integration**, enabling **real-time financial analysis** and **reducing monthly reporting time by 30%.**

Personal Projects

PageSpeaks — Multi-Voice TTS Platform |Try me!| *Python, PyTorch, F5-TTS, Whisper, Modal, FastAPI, Next.js* ***May* 2026**

- Fine-tuned **F5-TTS** (flow-matching voice synthesis) on **19 hours** of lecture audio, producing **6,676 training clips** and deploying on **serverless Modal GPU** with **HuggingFace Hub** model storage.
- Built an **end-to-end audio preprocessing pipeline** in **Python — silence-splitting**, **noise reduction**, and **Whisper-based transcription** converting raw MP3 lectures into a clean **LJSpeech-format** training dataset.
- Implemented **chunked streaming inference** on **serverless GPU**, **prefetching** the next audio segment while the current one plays to minimize perceived latency across long-form text.

AI-Powered Professor Rating Platform |Try me!| LangChain, FastAPI, Redis, WebSockets, JavaScript **August 2025**

- Built an AI-powered professor search platform using **LangChain + Pinecone (RAG)** on RateMyProfessor data, enabling students to discover and evaluate professors with significantly improved relevance over traditional keyword search.
- Designed a **FastAPI WebSocket** backend with **Redis caching** that streams token-by-token responses in real time, with an agentic RAG-to-GraphQL fallback pipeline ensuring consistent answer coverage across queries.
- Developed a responsive web interface with HTML/CSS featuring **AI-driven professor recommendations**, follow-up context handling, and side-by-side professor comparisons for an interactive search experience.

Certification

AWS Certified Solutions Architect – Associate (Credential ID: link) **Expires March 2029**